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A constrained multi-objective evolutionary algorithm based on fitness landscape indicator

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ABSTRACT

Constrained multi-objective optimization problems (CMOPs) with constraints in both the decision and objective space are shown to be great challenges to be solved. Considering the different requirements of different problems on resource allocation of exploration and exploitation, this paper proposes a new constrained multi-objective evolutionary algorithm based on a novel fitness landscape indicator. The indicator regards the fitness landscape and evolutionary generation among the population to determine the selection of the offspring generation mechanism. The proposed algorithm uses the new indicator to select different differential evolutions during the evolutionary process to balance exploration and exploitation. Numerical experiments on three test suites and three practical examples compared with six existing algorithms show the proposed algorithm can effectively deal with different types of CMOPs, especially in CMOPs with constraints in both the decision and objective spaces.

1. Introduction

The constrained multi-objective optimization problems (CMOPs) considered in this paper are generally formulated as follows:

$$\begin{cases} \text{minimize} & \mathbf{F}(\mathbf{x}) = \left(f_1(\mathbf{x}), \dots, f_m(\mathbf{x})\right)^T \\ \text{subject to} & g_i(\mathbf{x}) \geq 0, i = 1, \dots, q \\ & h_j(\mathbf{x}) = 0, j = 1, \dots, p \\ & \mathbf{x} \in \mathbb{R}^n \end{cases} \tag{1}$$

where $\mathbf{F}(\mathbf{x}) = \left(f_1(\mathbf{x}), f_2(\mathbf{x}), \dots, f_m(\mathbf{x})\right)^T \in \mathbb{R}^m$ is an m-dimensional objective vector, $g_i(\mathbf{x})$ is the ith inequality constraint, $h_j(\mathbf{x})$ is the jth equality constraint, q and p are the numbers of inequality and equality constraints, respectively, $\mathbf{x} \in \mathbb{R}^n$ is an n-dimensional decision vector. Since the equality constraints are difficult to solve directly, a very small positive number ϵ $\left(\text{e.g.}, \epsilon = 10^{-6}\right)$ is used to convert equality constraints into inequality constraints. The specific representation is given as follows :

$$h_j(\mathbf{x})' = \left|h_j(\mathbf{x})\right| - \epsilon \leqslant 0, \quad j = 1, \dots, p \tag{2}$$

Usually, the overall constraint violation of an individual $\mathbf{x}$ is calculated as:

$$G(\mathbf{x}) = \sum_{i=1}^{q} \max\left(g_i(\mathbf{x}), 0\right) + \sum_{j=1}^{p} \max\left(h_j(\mathbf{x})', 0\right) \tag{3}$$

Obviously, $G(\mathbf{x}) = 0$ when the individual $\mathbf{x}$ satisfies all the constraints. In this case, the individual $\mathbf{x}$ is called a feasible solution; otherwise, it is an infeasible solution. For a CMOP, the set of all feasible solutions is called the feasible region.

For any two feasible individuals $\mathbf{x}^1, \mathbf{x}^2 \in S$, if $f_i\left(\mathbf{x}^1\right) \leq f_i\left(\mathbf{x}^2\right)$ for all $i \in \{1, \dots, m\}$ and $f_j\left(\mathbf{x}^1\right) < f_j\left(\mathbf{x}^2\right)$ for at least one $j \in \{1, \dots, m\}$, then $\mathbf{x}^1$ is said to dominate $\mathbf{x}^2$, denoted as $\mathbf{x}^1 \prec \mathbf{x}^2$. If $\nexists \mathbf{x}^a \in S$ such that $\mathbf{x}^a \preceq \mathbf{x}^*$, then $\mathbf{x}^*$ is called the Pareto optimal solution of CMOPs. The set consisting of all Pareto optimal solutions is called the Pareto optimal set (PS). The image of the PS in the objective space is called a Pareto optimal front (PF), denoted as $PF = \{\mathbf{F}(\mathbf{x}) \mid \mathbf{x} \in PS\}$. A key issue in solving CMOPs is maintaining a balance between convergence and diversity. Therefore, the following two aspects must be satisfied: (1) the feasible solutions should be as close as possible to the real PF of the problem; (2) the approximate optimal solution set obtained should be uniformly distributed along PF.

Various problems in science and engineering disciplines often contain several conflicting objectives and multiple complex constraints. These problems can be formulated as CMOPs, such as the fine-grained power cap allocation for power-constrained systems [1], the large-scale design optimization of electric machines [2], the optimization of spare parts inventory [3], and the problem of resource provisioning and scheduling of scientific workflows in uncertain cloud environments [4]. CMOPs encountered in practical applications usually involve two types

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of constraints: (1) constraints in the decision space; and (2) constraints in the objective space. Besides, it is very difficult to solve CMOPs, which contain both the decision constraints and objective constraints [5].

Over the past decade, many constrained multi-objective evolutionary algorithms (CMOEAs) have been proposed [6]. These CMOEAs show their competitive performance in solving CMOPs [7–9]. According to the constraint handling techniques (CHTs), these CMOEAs can be roughly classified into four types. The first type is the penalty function method [10–13]. These methods mainly add a penalty factor to the function objectives. In this way, a constrained multi-objective optimization problem is converted into an unconstrained multi-objective optimization problem which can be effectively solved by a multi-objective optimization algorithm. Although these methods are easy to implement, the selection of penalty factor is problem-dependent and hard to determine. The second type of method separates the objectives and constraints [14,15]. Among these methods, representative algorithms include constraint dominance principle (CDP) [16], stochastic ranking (SR) [17], epsilon constraint-handling method (EC) [18]. As CDP is simple and parameter-free, it is embedded into the Non-dominated Sorting Genetic Algorithm (NSGA-II) [19] and the Multi-objective Evolutionary Algorithm Based on Decomposition (MOEA/D) [20] as NSGA-II-CDP and MOEA/D-CDP [21]. However, CDP states that feasible solutions are always better than infeasible solutions. It can easily lead to the population falling into local feasible regions [22]. The third type is multi-objective optimization methods [14,23,24]. The methods regard the constraints as additional optimization objectives and convert a CMOP into a multi-objective optimization problem (MOP), which can be solved by the unconstrained MOEAs. The last type is hybrid methods [25–28]. Among these methods, they can be further divided into multi-CHT-based CMOEAs, multi-phase-based CMOEAs, and multi-population-based CMOEAs. Multi-CHT-based methods combine two or more different constraint handling techniques to solve the CMOPs [27,28]. Multi-phase-based methods divide the search process into multiple phases. In each phase, different CHT is utilized according to the current state [9]. The multi-population-based CMOEAs usually use two populations to solve CMOP cooperatively [7,8,29]. The multi-population-based CMOEAs show their competitive performance in solving CMOPs.

While extensive numerical experiments show that existing CMOEAs exhibit excellent performance in solving CMOPs [6], only a few CMOEAs attempt to solve CMOPs with constraints in both the decision and objective spaces [5]. Two-phase Framework (Top) is a two-phased-based CMOEA to solve such problems. In the first phase, a simple weight is used to transform a CMOP into a constrained single-objective optimization problem. Then the information on the population obtained in the first phase can be utilized in the second phase. However, the simple weight is hard to select without prior knowledge. Therefore, how to design an effective algorithm for solving CMOPs with constraints in both the decision and objective spaces is still a challenging problem.

Differential evolution (DE) is a well-known heuristic optimization strategy [30]. For each selected individual, it generates a new individual by mutating it in the scale of the difference between it and other selected individuals. Recently, several works showed that the performance of different optimization problems is largely determined by the selection of the mutation strategy in DE [31–33]. These works presented that a careful selection of different DEs by the analysis of the fitness landscape of an optimization problem can improve performance compared with using a simple DE. For example, papers [31,32] choose different DEs based on the proportion of positive correlation between the distance from the individual in the population to the current optimal and its fitness value. Although these works have been successfully developed in single-objective and multi-objective optimization problems, they did not consider the CMOPs which have a more complex fitness landscape especially CMOPs with constraints in both the decision and objective spaces. This inspired us to design a constrained multi-objective evolutionary algorithm to deal with the COMPs with constraints in both the decision and objective spaces which uses a new fitness landscape-based indicator to carefully select different types of DE during the evolutionary process.

Keeping the above motivation in mind, this paper first proposes an indicator to determine which type of DE to use. Specifically, the indicator is defined based on the fitness distance correlation [34] and current evolution generation. In each generation, if the solution far from the optimal solution set has a better fitness, the DE that avoids premature convergence is preferred to be used; Otherwise, the DE that speeds up the rate of population convergence is more likely to be chosen. In this way, the population is prone to avoid falling into local regions. This paper further incorporates the indicator into a CMOEA, *i.e.*, Indicator-based Constrained Multi-objective Algorithm (ICMA) [29], to solve the CMOPs with constraints in both the decision and objective spaces. Experiments on several benchmark suites demonstrate that the proposed algorithm performs better than the other six state-of-the-art comparison algorithms in dealing with different types of CMOPs.

The rest of this paper is organized as follows. Section 2 gives the preliminary knowledge, including fitness landscape, DE, and ICMA. Section 3 presents the proposed indicator and the framework of the algorithm. Section 4 compares the proposed algorithm with six CMOEAs on a series of benchmarks and discusses the experimental results. Finally, in Section 5, the conclusion of this paper is drawn.

2. Preliminary knowledge

2.1. Differential evolution

In the field of evolutionary computation, DE shows its excellent performance in solving various complex optimization problems [35–37] and practical application problems [38–40]. In general, the existing DE can be mainly grouped into two categories, where the mutation strategies are different in these categories.

The first category randomly selects individuals in mutation operation. In this way, these algorithms can fully explore the whole search space and avoid premature convergence. Among them, DE/rand/2 is a popular algorithm to maintain the diversity of the population [41]. The second category focuses on the best individual in mutation operation. They can speed up the convergence by using information about the best individual found during the evolution. In DE/Best/1, the best individual is considered as a center in mutation operation which leads to rapid convergence [42]. The first category is useful in solving multi-model problems while the second one is suitable in dealing with uni-model problems due to their different characteristics [30].

2.2. Fitness landscape

Different categories of DE are suitable for different optimization problems. Since the prior knowledge of an optimization problem is generally unknown, the fitness landscape is used to measure its characteristic [43,44]. The fitness landscape is constituted by the fitness values of all individuals. As shown in paper [45], the fitness landscape with the different distributions, shapes, and numbers of modes can affect the difficulty of solving optimization problems. From the perspective of the fitness landscape, paper [46] demonstrates that a search strategy related to the characteristics of the fitness landscape is conducive to tackling optimization problems effectively.

Recently, many fitness landscape analysis techniques have been developed and used to guide the design of the optimization algorithm [47]. Paper [48] proposes a self-feedback differential evolution algorithm that adapts to the fitness landscape to solve single-optimization problems. It measures the fitness landscape by calculating the number of individuals in the population whose distance to the optimal is positively correlated with their objective value. Paper [49] also proposes a mixed strategy for different landscape features by using the above fitness landscape. However, this fitness landscape measurement fails to consider the nonlinear among the solutions. Paper [50] uses a popular fitness distance correlation (FDC) [34] to adaptive select mutation strategy. The FDC measures the characteristics of the fitness landscape by quantifying the correlation between fitness values υ and distances d. Given a set of n individuals $\mathbf{X}=\{\mathbf{x}_1,\dots,\mathbf{x}_n\}$, the FDC value r_{VD} can be computed as follows:

$$r_{VD}=\frac{C_{VD}}{S_V S_D} \tag{4}$$

$$C_{VD}=\frac{1}{n}\sum_{i=1}^{n}\left(\upsilon_i-\bar{\upsilon}\right)\left(d_i-\bar{d}\right) \tag{5}$$

where υ_i and d_i are the fitness value and the distance from the best individual of ith individual, respectively; $\bar{\upsilon}$ and $\bar{d}$ are the mean values of fitness and distance, respectively; S_V and S_D are the standard deviation of fitness and distance, respectively.

Different from dealing with single-objective optimization problems, some papers focus on constrained optimization problems [51,52] and multi-objective optimization problems [53,54] by analyzing the fitness landscape. Paper [52] measures the difference between the fitness landscape of a problem and a reference fitness landscape to guide the selection of DE to deal with constrained optimization problems. Paper [54] considers the hypervolume indicator as the fitness value to analyze the fitness landscape of a multi-objective optimization problem. These works have not considered the CMOPs. Also, they use the indicator based on the fitness landscape to select different optimization algorithms which fail to consider that different stages of the optimization process should prefer different degrees of exploration.

2.3. Indicator-based constrained multi-objective algorithm

The main idea of the ICMA is to use an indicator to guide the population uniformly to explore promising regions. It can prevent the population from getting stuck in local regions. In ICMA, the novel indicator is defined by constraints and a cost-valued distance. This indicator can evaluate the quality of feasible and infeasible solutions in the population.

Given a set of n individuals $\mathbf{X}=\{\mathbf{x}_1,\dots,\mathbf{x}_n\}$, the fitness value FV_i of an individual $\mathbf{x}^i$ can be calculated as follows:

$$FV_i=I\left(\mathbf{x}^i\mid\mathbf{X}\right)=\min_{\mathbf{x}^j\in\mathbf{X}\setminus\mathbf{x}^i} I\left(\mathbf{x}^i\mid\mathbf{x}^j\right) \tag{6}$$

where $I\left(\mathbf{x}^i\mid\mathbf{x}^j\right)$ is a function that evaluates the individual $\mathbf{x}^i$ by the individual $\mathbf{x}^i$. Especially,

$$I\left(\mathbf{x}^i\mid\mathbf{x}^j\right)=\begin{cases}\log CV\left(\mathbf{y}^i\mid\mathbf{y}^j\right), & \text{if } G(x^i)=0, G(x^j)=0\\ \infty, & \text{if } G(x^i)=0, G(x^j)\neq 0\\ \log CV\left(\mathbf{y}^i\mid\mathbf{y}^j\right), & \text{if } G(x^i)\neq 0, G(x^j)=0\\ \log\max(c^{CV}(\mathbf{y^i},\mathbf{y^j}),\frac{G(x^j)}{G(x^i)}), & \text{otherwise}\end{cases} \tag{7}$$

where $\mathbf{y}^i=\mathbf{F}(\mathbf{x^i})$ and $c^{CV}(\mathbf{y^i},\mathbf{y^j})=\max(CV\left(\mathbf{y}^i\mid\mathbf{y}^j\right),CV\left(\mathbf{y}^j\mid\mathbf{y}^i\right))$, the symbol CV is an improved ratio-based cost value computed as follows:

$$CV\left(\mathbf{y}^i\mid\mathbf{y}^j\right)=\begin{cases}\max_{p\in\{1,\dots,m\}}\frac{f_p^j}{f_p^i}, & \text{if } \exists f_p^j>f_p^i;\\ \min_{p\in\{1,\dots,m\}}\frac{f_p^j}{f_p^i}, & \text{otherwise.}\end{cases} \tag{8}$$

An individual with a larger fitness value means better quality.

3. Proposed algorithm

3.1. Proposed indicator

This paper proposes an indicator that uses the fitness distance correlation and the evolutionary generation to measure the selection of DE. The indicator $I_t(\mathbf{X})$ for a population $\mathbf{X}$ at tth generation is defined as follows:

$$I_t(\mathbf{X})=(\frac{r_{FD}-\min(Ran)}{\max(Ran)-\min(Ran)})^{\alpha}*\frac{t}{T} \tag{9}$$

where r_{VD} is the fitness distance correlation of population X; Ran is the range of r_{FD}, $\max(Ran)$ and $\min(Ran)$ are the maximum and minimum of the range Ran, respectively; the α is a nonlinear parameter with range of $[0,\infty)$ and T is the maximum of evolutionary generation.

In this paper, the proposed indicator is embedded into ICMA. In such a way, the FV_i in Eq. (5) is set as the fitness value. Due to the optimal of a CMOP involving several individuals, this paper uses a set of optimal individuals to calculate r_{VD}. In the proposed indicator, there are two characteristics below:

- A larger fitness distance correlation leads to a larger indicator value; otherwise, a smaller fitness distance correlation brings a smaller indicator value.
- With the increase of evolutionary generation, the indicator gradually grows. At the early stages of evolution, the indicator tends to be zeros. At the late stage of evolution, the indicator is similar to the normalizing r_P.

A small value may be caused by the evolutionary generation being in the early stage or/and a small fitness distance correlation. Therefore, when the value of the indicator is small, an algorithm should focus on exploring other areas; otherwise, it should tend to exploit the current area.

Algorithm 1 General framework of the proposed algorithm.

Input:

- N : the population size N;
- t : the current generation index;
- T : the maximum generation.

Output: *Archive* Λ: a set of feasible non-dominated solutions.

```
Initialization:
t=1; initialize an population X = {x_1, ..., x_N} and let archive Λ = X;
while t < T do
    /* Step I: Calculate the proposed indicator I_t(X). */
    if mod(t,100)==0 then
        Calculate the fitness value of each individual in the population by Eq. (5);
        Calculate the indicator I_t(X) by Eq. (9);
    end if
    /* Step II: Offspring Production. */
    if rand ⩽ I_t(X) then
        Use DE/best/1 to generate offspring population C;
    else
        Use DE/rand/2 to generate offspring population C.
    end if
    /* Step III: Population and Archive Update. */
    MaxPop = X ∪ Λ ∪ C;
    Update population X according to the FV;
    Update archive Λ according to the constraint violation;
    t = t + 1;
end while
```

3.2. Proposed framework

Since there are massive local regions in a CMOP with constraints in both the decision and objective spaces, it leads the population easily to fall into a local optimal region. This paper uses to proposed indicator to measure the fitness landscape. By embedding the proposed indicator into ICMA and measuring the fitness landscape, this paper

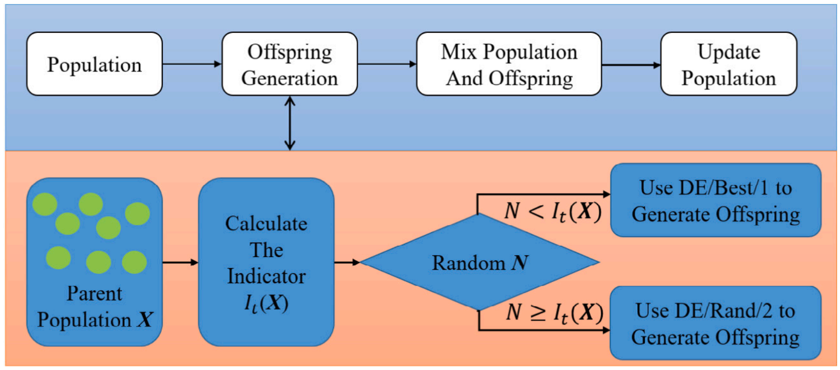

Fig. 1. The diagram of the proposed algorithm framework.

Table 1
The general information of three artificial examples.

Test instance	m	Characteristics	r_{FD}
F1	2	constraints in decision space and objective space decrease as the distance from PS decreases, respectively	$7.89E^{-1}$
F2	2	constraints in decision space and objective space are independent of the distance from PS, respectively	$5.94E^{-4}$
F3	2	constraints in decision space and objective space decrease as the distance from PS, respectively	$-1.89E^{-1}$

uses different types of DE to generate an offspring population. The proposed framework of the constrained multi-objective optimization algorithm is shown in algorithm 3.1. It contains two populations: working population $\mathbf{X}$ and archive population Λ. In each generation, the proposed algorithm can be mainly divided into three steps. In the first step, it calculates the indicator $I_t(\mathbf{X})$ for the population $\mathbf{X}$ according to the fitness value of each individual in the population. In the second step, the offspring population $\mathbf{C}$ is generated by using specific DE based on the indicator $I_t(\mathbf{X})$. When the indicator is large, the proposed algorithm tends to use DE/best/1 to generate the offspring. In this way, the offspring population prefers exploiting the current search areas. Otherwise, more offspring individuals will be generated randomly by using DE/rand/2. In the final step, the working population and archive are updated according to the fitness value and constraint violation, respectively. Repeat the above loop and final output all feasible solutions in archive Λ. The diagram of the proposed algorithm framework is given in Fig. 1.

4. Experimental studies

4.1. Different DE selection in evolution

This paper set up three artificial examples to show the different DE selections during the evolution process in this subsection. The detail formula of each problem is shown in the appendix. The characteristics of constraints for these problems are shown in Table 1. In addition, Table 1 shows the FDC value r_{FD} for each problem. This paper shows the relationship between fitness value and distance in the scatter plot for each problem, as Fig. 2.

Table 1 shows the relationship between constraints and the distance from PF for each example. For the F1, constraints in decision space and objective space have a positive correlation with the distance from PS. For this example, more exploitation should be emphasized due to the decrease in constraint is consistent with convergence. For F2, the constraints are independent of the distance from PS. In this example, both exploration and exploitation should be considered due to many local regions. As to the F3, the change of constraints has a negative correlation with distance from PS. In this situation, the population is easy to fall into the local region away from the PS. Therefore, it requires paying more attention to exploring other areas. As shown in Fig. 2, the fitness value is a positive correlation with distance from the PS for F1. In addition, As shown in Table 1, the FDC value r_{FD} is close to 1, which is $7.89E^{-1}$. In terms of the F2, the FDC value r_{FD} is close to 0. The fitness value is random to the distance. As for the F3, the FDC value r_{FD} is a negative number. The fitness value decreases slightly when the distance increases. Therefore, the FDC value r_{FD} can reveal whether to prefer exploration or exploitation. A larger r_{FG} means more exploitation should be considered; otherwise, more exploration should be interested.

Fig. 3 shows the fitness value for each example in three stages, *i.e.*, early stage, mid-stage, and later stage. As shown in Fig. 3, all indicator values increase when the evolutionary generation grows. Therefore, the DE/rand/2 will be preferred in the early stage. At the later stage of evolution, more resources are used to search for the current optimal solutions. Compared with the three examples, the proposed algorithm tends to select DE/rand/2 for the example with complex constraints. *i.e.*, F2 and F3, during the whole evolutionary process. In contrast, the DE/best/1 is more inclined to be used for solving an example whose change of constraint violation is consistent with convergence.

4.2. Experiments of test suites

4.2.1. Comparison algorithms

This paper selected six the-state-of-art CMOEAs, *i.e.*, Indicator-based Constrained Multi-objective Algorithm (ICMA) [29], Two-phase Framework (ToP) [5], Two-archive Evolutionary Algorithm (C-TAEA) [7], Detect-and-Escape Strategy (MOEA/D-DAE) [55], Push and Pull Search (PPS) [9] and Tri-goal Evolutionary Framework (TiGE$_2$) [56] for the performance comparison. The general ideas of the comparison algorithms are given as follows:

- ICMA [29] is an indicator-based constrained multi-objective optimization algorithm. It proposes a new indicator to guide the population to search uniformly over the promising area. ICMA is shown to deal effectively with the complex problems of irregular PFs.

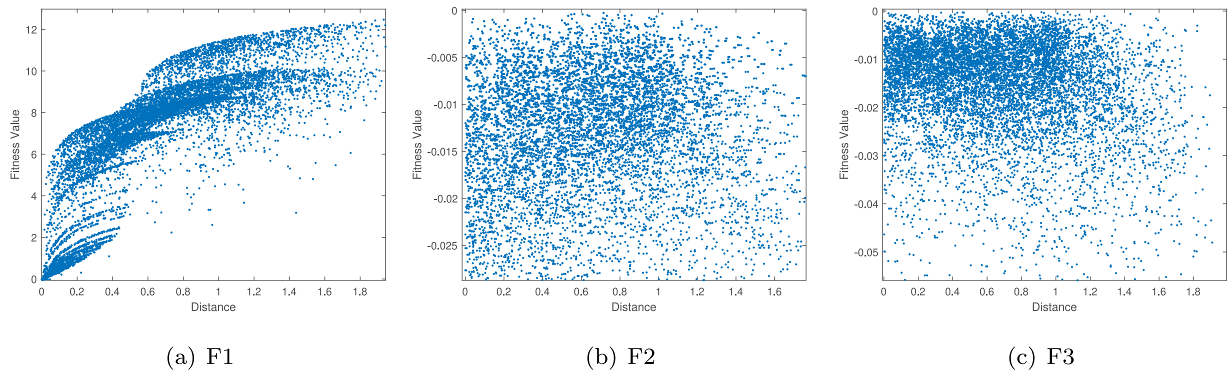

Fig. 2. Scatter plot of fitness f versus distance d to the optimal solution set.

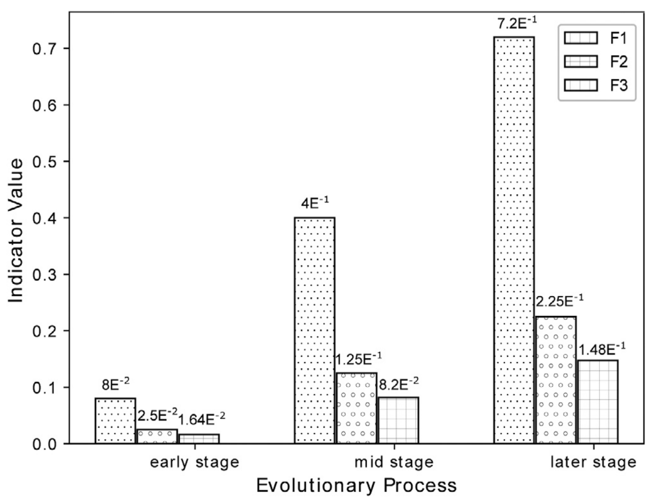

Fig. 3. The proposed indicator value for each example during the evolutionary process.

- ToP [5] is a two-stage constrained multi-objective evolutionary algorithm. In the first stage, a CMOP is transformed into a constrained single objective optimization problem and solved by a constrained optimization algorithm. In the second stage, the existing CMOEA is applied directly to deal with the CMOP.
- C-TAEA [7] is a co-evolutionary algorithm for solving CMOPs. It maintains two populations during the evolutionary process. By balancing the diversity and feasibility of each population, respectively, it is suitable to solve various CMOPs.
- MOEA/D-DAE [55] is a CMOEA with a detect-and-escape strategy. The strategy can help the population jump out of local optima. When the search space is divided into several discontinuous feasible domains by constraints, it is easy for the population to fall into local feasible or infeasible regions. To address this issue, MOEA/D-DAE detects whether the current population falls into local regions by the change rates of feasible solutions and constraint violation degree. After that, MOEA/D-DAE escapes the population from the local regions by adjusting the weights of constraint violations.
- PPS [9] is a two-stage CMOEA that adopts a push-and-pull strategy. In the pushing stage, PPS does not consider the constraints that make the population cross the infeasible region. In the pull stage, it considers constraints to facilitate pulling the population back to the PF.
- TiGE$_2$ [56] is a tri-goal evolution algorithm for solving CMOPs. It designs three indicators for diversity, convergence, and feasibility, respectively. The algorithm is shown as conceptually simple.

4.2.2. Test suites and parameter settings

Three test suites, *i.e.*, CF [57], DOC [5], and MW [58], were used to examine the performance of the proposed algorithm. Both the CF suite and DOC suite contain constraints in the decision and objective spaces. The MW suite considers the constraints in the objective space.

Each algorithm was run 20 times independently on each test problem. The size of the population and the maximum number of function evaluations of each algorithm on each test problem are set at 100 and $5E^5$, respectively. The private parameter settings of each comparison algorithm are consistent with their original papers, respectively. All the comparison algorithms are run on the evolutionary multi-objective optimization platform (PlatEMO) [59] which was developed on Matlab. The source codes of the proposed algorithm are available at https://github.com/jjfang123/FLI.

4.2.3. Performance metrics

To compare the performance of the seven constraint algorithms, two common metrics were adopted in our experiments, *i.e.*, inverted generational distance (IGD) [60] and hypervolume(HV) [61].

- Given P^* as the set of points uniformly distributed on the real PF, and X is a set of feasible optimal solutions obtained by an algorithm. The IGD value of X is calculated as:

$$IGD(P^*, X) = \frac{\sum_{v \in P^*} d(v, X)}{|P^*|}. \tag{10}$$

where $d(v, X)$ is the minimum Euclidean distance from individual v to population X in P^*, and $|P^*|$ is the size of P^*. The smaller the value of the IGD, the better the convergence and diversity performance of the algorithm. The set P^* with a size of 10,000 is generated by the method of Das and Dennis [62] for computing IGD.
- HV measures the volume of the region in the objective space bounded by the feasible solution set X obtained by an algorithm and the reference point z.

$$HV(X \mid \mathbf{z}) = VOL\left(\bigcup_{\mathbf{x} \in X} \left[f_1(\mathbf{x}), z_1\right] \times \cdots \times \left[f_m(\mathbf{x}), z_m\right]\right) \tag{11}$$

where $VOL(*)$ is the Lebesgue measure, which is used to measure volume. The reference point z is composed of 1.1 times the maximum function value for each objective on PF. A larger HV value indicates a better performance of the algorithm.

4.2.4. Numerical results of test suites

The statistical results of the average IGD and HV on CF1-16 and DOC1-9 achieved by seven CMOEAs over 20 independent runs are listed in Table 2 and Table 3, respectively. A total of 28 instances were tested. In Table 2 and Table 3, the symbol “NaN” means the corresponding algorithm fails to find any feasible solution in all 20

Table 2
The average values of IGD metric of the final solutions obtained by all algorithms over 20 independent runs on CF and DOC test suites.

Problem	ICMA	ToP	C-TAEA	MOEA/D-DAE	PPS	$TiGE_2$	Proposed algorithm
CF1	$4.64E^{-2}$ –	$8.78E^{-2}$ –	$4.67E^{-2}$ –	$4.72E^{-2}$ –	$5.44E^{-2}$ –	$1.12E^{-1}$ –	$4.52E^{-2}$
CF2	$6.08E^{-2}$ –	$1.37E^{-1}$ –	$1.31E^{-1}$ –	$1.87E^{-1}$ =	$7.10E^{-2}$ –	$1.90E^{-1}$ –	$5.94E^{-2}$
CF3	$6.16E^{-2}$ –	$1.53E^{-1}$ –	$7.64E^{-2}$ –	$6.59E^{-2}$ –	$7.29E^{-2}$ –	$1.23E^{-1}$ –	$6.04E^{-2}$
CF4	$1.47E^{-1}$ +	$5.44E^{-1}$ –	$1.24E^{-1}$ +	$2.91E^{-1}$ –	$5.15E^{-1}$ –	$3.02E^{-1}$ –	$1.51E^{-1}$
CF5	$2.97E^{-2}$ –	$6.05E^{-2}$ –	$3.62E^{-2}$ –	$7.54E^{-2}$ –	$4.93E^{-2}$ –	$6.45E^{-2}$ –	$2.94E^{-2}$
CF6	$3.07E^{-2}$ =	$5.35E^{-2}$ –	$3.67E^{-2}$ –	$3.21E^{-2}$ –	$4.81E^{-2}$ –	$5.80E^{-2}$ –	$3.08E^{-2}$
CF7	$1.07E^{-1}$ –	$3.69E^{-1}$ –	$8.76E^{-2}$ –	$1.84E^{-1}$ –	$3.03E^{-1}$ –	$2.12E^{-1}$ –	$4.48E^{-2}$
CF8	$3.50E^{-2}$ –	$1.67E^{-1}$ –	$4.38E^{-2}$ –	$3.41E^{-2}$ –	$5.57E^{-2}$ –	$7.46E^{-2}$ –	$3.31E^{-2}$
CF8_II	$4.59E^{-2}$ =	NaN-	$6.69E^{-2}$ –	$3.36E^{-2}$ =	$6.38E^{-2}$ –	$1.61E^{-1}$ –	$4.70E^{-2}$
CF9	$3.11E^{-2}$ –	$4.09E^{-1}$ –	$6.58E^{-2}$ –	$3.16E^{-2}$ –	$5.96E^{-2}$ –	$9.90E^{-2}$ –	$3.05E^{-2}$
CF9_II	$3.95E^{-2}$ +	NaN-	$6.65E^{-2}$ =	$3.26E^{-2}$ +	$1.08E^{-1}$ =	$1.04E^{-1}$ =	$1.44E^{-1}$
CF10	$1.31E^{-1}$ –	$8.10E^{-1}$ –	$1.16E^{-1}$ –	$2.84E^{-1}$ –	$5.58E^{-1}$ –	$1.66E^{-1}$ –	$5.47E^{-2}$
CF11	$2.95E^{-2}$ =	$4.30E^{-1}$ –	$3.62E^{-2}$ –	$3.06E^{-2}$ –	$3.91E^{-2}$ –	$1.64E^{-1}$ –	$2.95E^{-2}$
CF12	$2.52E^{-2}$ –	$3.83E^{-1}$ –	$1.07E^{-1}$ –	$8.94E^{-2}$ –	$6.16E^{-2}$ –	$1.19E^{-1}$ –	$2.48E^{-2}$
CF13	$1.62E^{-1}$ –	$7.18E^{-1}$ –	$1.43E^{-1}$ –	$5.88E^{-1}$ –	$3.83E^{-1}$ –	$3.07E^{-1}$ –	$7.13E^{-2}$
CF14	$3.51E^{-2}$ –	$2.06E^{-1}$ –	$4.09E^{-2}$ –	$3.71E^{-2}$ –	$5.25E^{-2}$ –	$1.13E^{-1}$ –	$3.44E^{-2}$
CF14_II	$3.81E^{-2}$ +	NaN –	$4.09E^{-2}$ +	$3.59E^{-2}$ +	$3.26E^{-1}$ –	$1.83E^{-1}$ –	$7.90E^{-2}$
CF15	$4.25E^{-2}$ =	$3.02E^{-1}$ –	$5.64E^{-2}$ –	$2.32E^{-1}$ –	$6.95E^{-2}$ –	$1.49E^{-1}$ –	$4.20E^{-2}$
CF16	$4.29E^{-2}$ =	$1.74E^{-1}$ –	$5.60E^{-2}$ –	$4.68E^{-2}$ –	$6.57E^{-2}$ –	$1.21E^{-1}$ –	$4.29E^{-2}$
DOC1	7.01 –	$5.83E^{-3}$ +	$4.20E^{2}$ –	$1.60E^{-1}$ –	$5.92E^{-2}$ –	1.75 –	$3.11E^{-2}$
DOC2	$4.24E^{-2}$ –	$3.11E^{-1}$ –	NaN –	$8.74E^{-2}$ =	$3.93E^{-1}$ –	NaN –	$1.24E^{-2}$
DOC3	$9.17E^{2}$ –	$4.44E^{1}$ +	NaN –	$2.01E^{2}$ +	$1.97E^{2}$ +	$5.90E^{2}$ =	$5.14E^{2}$
DOC4	5.38 –	$3.75E^{-2}$ +	$1.53E^{2}$ –	$7.21E^{-1}$ –	$2.82E^{-1}$ –	1.46 –	$1.38E^{-1}$
DOC5	$3.70E^{1}$ =	$4.74E^{1}$ =	NaN –	NaN –	$6.86E^{1}$ –	$9.01E^{1}$ –	$5.28E^{1}$
DOC6	7.46 –	$9.40E^{-1}$ –	$3.04E^{1}$ –	$4.98E^{-1}$ –	$5.25E^{-1}$ –	$9.01E^{-1}$ –	$4.25E^{-1}$
DOC7	6.46 –	$6.72E^{-2}$ +	NaN –	$6.44E^{-1}$ +	$3.86E^{-1}$ +	3.67 =	4.73
DOC8	8.02 –	3.46 –	$3.28E^{2}$ –	$3.42E^{1}$ –	$7.39E^{1}$ –	$1.39E^{2}$ –	$5.59E^{-2}$
DOC9	$1.01E^{-1}$ +	$1.43E^{-1}$ +	$8.20E^{-1}$ –	$2.78E^{-1}$ –	$2.55E^{-1}$ –	$6.50E^{-2}$ +	$1.54E^{-1}$
+/−/=	4/18/6	5/22/1	2/25/1	4/21/3	2/25/1	1/24/3	

The symbols "−", "=", and "+" denote that the proposed algorithm is better than, similar to and inferior to the compared algorithm under the Wilcoxon's rank sum test with a 0.05 significance level. The best result in each row is highlighted.

Table 3
The average values of HV metric of the final solutions obtained by all algorithms over 20 independent runs on CF and DOC test suites.

Problem	ICMA	ToP	C-TAEA	MOEA/D-DAE	PPS	$TiGE_2$	Proposed algorithm
CF1	$8.30E^{-1}$ –	$7.67E^{-1}$ –	$8.33E^{-1}$ +	$8.32E^{-1}$ =	$8.08E^{-1}$ –	$7.61E^{-1}$ –	$8.32E^{-1}$
CF2	$5.48E^{-1}$ –	$3.97E^{-1}$ –	$5.08E^{-1}$ –	$4.76E^{-1}$ –	$5.20E^{-1}$ –	$4.52E^{-1}$ –	$5.52E^{-1}$
CF3	$5.15E^{-1}$ –	$3.48E^{-1}$ –	$5.11E^{-1}$ –	$5.20E^{-1}$ +	$4.98E^{-1}$ –	$4.54E^{-1}$ –	$5.18E^{-1}$
CF4	$3.15E^{-1}$ –	$8.30E^{-2}$ –	$3.29E^{-1}$ +	$1.98E^{-1}$ –	$1.26E^{-1}$ –	$1.85E^{-1}$ –	$3.25E^{-1}$
CF5	$3.16E^{-1}$ –	$2.68E^{-1}$ –	$3.12E^{-1}$ –	$3.00E^{-1}$ –	$3.03E^{-1}$ –	$2.70E^{-1}$ –	$3.17E^{-1}$
CF6	$2.31E^{-1}$ –	$1.99E^{-1}$ –	$2.27E^{-1}$ –	$2.31E^{-1}$ =	$2.23E^{-1}$ –	$2.06E^{-1}$ –	$2.31E^{-1}$
CF7	$2.58E^{-1}$ –	$9.06E^{-2}$ –	$2.81E^{-1}$ –	$1.91E^{-1}$ –	$1.65E^{-1}$ –	$1.42E^{-1}$ –	$3.43E^{-1}$
CF8	$2.31E^{-1}$ –	$1.53E^{-1}$ –	$2.31E^{-1}$ =	$2.39E^{-1}$ +	$2.29E^{-1}$ –	$2.17E^{-1}$ –	$2.33E^{-1}$
CF8_II	$2.19E^{-1}$ =	NaN –	$2.08E^{-1}$ –	$2.39E^{-1}$ +	$2.17E^{-1}$ +	$1.61E^{-1}$ –	$2.16E^{-1}$
CF9	$2.48E^{-1}$ =	$1.27E^{-1}$ –	$2.40E^{-1}$ –	$2.71E^{-1}$ +	$2.52E^{-1}$ +	$2.48E^{-1}$ =	$2.49E^{-1}$
CF9_II	$2.29E^{-1}$ +	NaN –	$2.40E^{-1}$ +	$2.71E^{-1}$ +	$2.15E^{-1}$ +	$2.42E^{-1}$ +	$1.27E^{-1}$
CF10	$1.88E^{-1}$ –	$4.64E^{-2}$ –	$2.02E^{-1}$ –	$1.39E^{-1}$ –	$8.60E^{-2}$ –	$2.13E^{-1}$ =	$2.80E^{-1}$
CF11	$4.04E^{-1}$ =	$2.32E^{-1}$ –	$3.91E^{-1}$ –	$4.25E^{-1}$ +	$4.16E^{-1}$ +	$2.98E^{-1}$ –	$4.05E^{-1}$
CF12	$7.31E^{-1}$ =	$4.32E^{-1}$ –	$6.66E^{-1}$ –	$7.02E^{-1}$ –	$7.17E^{-1}$ =	$5.98E^{-1}$ –	$7.31E^{-1}$
CF13	$3.74E^{-1}$ –	$6.41E^{-2}$ –	$4.29E^{-1}$ –	$1.97E^{-1}$ –	$2.80E^{-1}$ –	$2.20E^{-1}$ –	$4.94E^{-1}$
CF14	$8.10E^{-1}$ =	$6.19E^{-1}$ –	$8.12E^{-1}$ +	$8.19E^{-1}$ +	$7.84E^{-1}$ –	$7.17E^{-1}$ –	$8.10E^{-1}$
CF14_II	$8.04E^{-1}$ +	NaN –	$8.12E^{-1}$ +	$8.20E^{-1}$ +	$4.14E^{-1}$ –	$6.43E^{-1}$ –	$7.50E^{-1}$
CF15	$3.90E^{-1}$ =	$1.98E^{-1}$ –	$3.83E^{-1}$ –	$3.17E^{-1}$ –	$3.75E^{-1}$ –	$3.06E^{-1}$ –	$3.90E^{-1}$
CF16	$4.90E^{-1}$ =	$4.05E^{-1}$ –	$4.86E^{-1}$ –	$4.95E^{-1}$ +	$4.75E^{-1}$ –	$4.48E^{-1}$ –	$4.91E^{-1}$
DOC1	$2.04E^{-1}$ –	$3.45E^{-1}$ +	0.00 –	$2.37E^{-1}$ –	$2.93E^{-1}$ –	$2.16E^{-2}$ –	$3.37E^{-1}$
DOC2	$5.74E^{-1}$ –	$4.07E^{-1}$ –	NaN –	$5.07E^{-1}$ =	$2.46E^{-1}$ –	NaN –	$6.09E^{-1}$
DOC3	0.00 =	$7.47E^{-2}$ +	NaN –	0.00 =	$5.32E^{-3}$ =	0.00 =	0.00
DOC4	$1.64E^{-1}$ –	$5.20E^{-1}$ +	0.00 –	$3.36E^{-2}$ –	$2.49E^{-1}$ –	0.00 –	$4.76E^{-1}$
DOC5	$2.61E^{-1}$ =	$2.5E^{-1}$ =	NaN-	NaN –	$3.41E^{-2}$ –	$9.36E^{-5}$ –	$2.44E^{-1}$
DOC6	$9.06E^{-3}$ –	$2.15E^{-2}$ –	0.00 –	$1.22E^{-1}$ –	$1.63E^{-1}$ –	$7.44E^{-2}$ –	$3.84E^{-1}$
DOC7	0.00 =	$4.21E^{-1}$ +	NaN –	$1.02E^{-1}$ +	$1.28E^{-1}$ +	$6.67E^{-3}$ =	0.00
DOC8	$1.98E^{-1}$ –	0.00 –	0.00 –	0.00 –	0.00 –	0.00 –	$8.05E^{-1}$
DOC9	$2.08E^{-1}$ –	$6.48E^{-2}$ –	0.00 –	$8.67E^{-2}$ –	$4.08E^{-2}$ –	$2.59E^{-1}$ +	$1.94E^{-1}$
+/−/=	2/16/10	4/23/1	5/22/1	10/14/4	5/21/2	2/22/4	

The symbols "−", "=", and "+" denote that the proposed algorithm is better than, similar to and inferior to the compared algorithm under the Wilcoxon's rank sum test with a 0.05 significance level. The best result in each row is highlighted.

times runs. As shown in Table 2, the proposed algorithm achieves the best results on 17 test problems in terms of IGD value. As to the HV value, the proposed algorithm gets the best results on 8 test problems. In addition, it can be observed that the proposed algorithm significantly outperforms ICMA, ToP, C-TAEA, MOEA/D-DAE, PPS, and $TiGE_2$ on 13, 22, 22, 14, 21, and 24 problems of 28 test instances in terms of IGD and HV values, respectively. This phenomenon shows the effectiveness of the proposed algorithm on these test problems.

Compared with ICMA, the proposed algorithm uses different DEs during the evolutionary process. It leads to better performance on various test problems with constraints in both the decision and objective spaces. When the fitness landscape of a test problem is complex, the

Table 4
The average values of IGD metric of the final solutions obtained by all algorithms over 20 independent runs on MW test suite.

Problem	ICMA	ToP	C-TAEA	MOEA/D-DAE	PPS	$TiGE_2$	Proposed algorithm
MW1	3.08E^{-2} =	9.11E^{-2} –	6.45E^{-3} –	1.85E^{-3} –	2.61E^{-3} –	5.15E^{-1} –	1.78E^{-3}
MW2	6.95E^{-3} =	1.31E^{-1} –	2.21E^{-2} –	9.87E^{-2} –	1.15E^{-1} –	5.99E^{-1} –	4.74E^{-3}
MW3	4.82E^{-3} =	4.07E^{-1} –	4.50E^{-3} +	4.83E^{-3} –	6.69E^{-3} –	2.75E^{-2} –	4.70E^{-3}
MW4	4.43E^{-2} =	1.04E^{-1} –	4.71E^{-2} –	4.32E^{-2} +	5.45E^{-2} –	1.08E^{-1} –	4.41E^{-2}
MW5	2.99E^{-4} =	7.12E^{-1} –	4.12E^{-3} –	5.40E^{-5} +	1.89E^{-1} –	5.43E^{-1} –	3.06E^{-4}
MW6	2.89E^{-3} =	6.11E^{-1} –	1.21E^{-2} –	2.66E^{-1} –	4.52E^{-1} –	8.48E^{-1} –	2.92E^{-3}
MW7	4.45E^{-3} +	8.06E^{-3} –	5.58E^{-3} –	4.37E^{-3} +	5.76E^{-3} –	9.97E^{-2} –	4.61E^{-3}
MW8	4.36E^{-2} =	1.64E^{-1} –	5.50E^{-2} –	1.05E^{-1} –	1.30E^{-1} –	7.16E^{-1} –	4.31E^{-2}
MW9	4.83E^{-3} +	4.06E^{-1} –	7.91E^{-3} –	4.29E^{-3} +	7.81E^{-3} –	3.49E^{-1} –	5.56E^{-3}
MW10	3.64E^{-3} =	3.97E^{-1} –	1.68E^{-2} –	3.00E^{-1} –	3.67E^{-1} –	1.58E^{-1} –	3.62E^{-3}
MW11	6.25E^{-3} =	8.06E^{-3} –	1.12E^{-2} –	6.28E^{-3} =	7.50E^{-3} –	4.08E^{-2} –	6.37E^{-3}
MW12	4.94E^{-3} =	3.32E^{-1} –	7.44E^{-3} –	5.21E^{-3} –	7.37E^{-3} –	3.41E^{-2} –	5.01E^{-3}
MW13	1.10E^{-2} =	5.50E^{-1} –	3.46E^{-2} –	2.46E^{-1} –	3.11E^{-1} –	2.32 –	1.12E^{-2}
MW14	9.70E^{-2} =	1.77E^{-1} –	1.14E^{-1} –	1.35E^{-1} –	1.43E^{-1} –	1.63E^{-1} –	9.58E^{-2}
+/–/=	2/0/12	0/14/0	1/13/0	4/9/1	0/14/0	0/14/0	

The symbols "–", "=", and "+" denote that the proposed algorithm is better than, similar to and inferior to the compared algorithm under the Wilcoxons rank sum test with a 0.05 significance level. The best result in each row is highlighted.

Table 5
The average values of HV metric of the final solutions obtained by all algorithms over 20 independent runs on MW test suite.

Problem	ICMA	ToP	C-TAEA	MOEA/D-DAE	PPS	$TiGE_2$	Proposed algorithm
MW1	4.64E^{-1} =	4.06E^{-1} –	4.83E^{-1} =	4.90E^{-1} +	4.89E^{-1} +	1.08E^{-1} –	4.89E^{-1}
MW2	5.77E^{-1} =	4.15E^{-1} –	5.51E^{-1} –	4.51E^{-1} –	4.31E^{-1} –	1.50E^{-1} –	5.81E^{-1}
MW3	5.45E^{-1} =	3.03E^{-1} –	5.46E^{-1} +	5.45E^{-1} –	5.44E^{-1} –	5.27E^{-1} –	5.45E^{-1}
MW4	8.38E^{-1} =	7.52E^{-1} –	8.38E^{-1} –	8.40E^{-1} +	8.23E^{-1} –	7.70E^{-1} –	8.39E^{-1}
MW5	3.25E^{-1} =	9.55E^{-2} –	3.22E^{-1} –	3.25E^{-1} +	2.64E^{-1} –	1.52E^{-1} –	3.25E^{-1}
MW6	3.27E^{-1} =	1.16E^{-1} –	3.12E^{-1} –	1.89E^{-1} –	1.25E^{-1} –	8.62E^{-2} –	3.27E^{-1}
MW7	4.11E^{-1} =	4.06E^{-1} –	4.10E^{-1} –	4.13E^{-1} +	4.13E^{-1} +	3.43E^{-1} –	4.11E^{-1}
MW8	5.48E^{-1} =	3.75E^{-1} –	5.21E^{-1} –	4.12E^{-1} –	3.56E^{-1} –	1.29E^{-1} –	5.48E^{-1}
MW9	3.97E^{-1} +	1.63E^{-1} –	3.95E^{-1} =	4.01E^{-1} +	3.91E^{-1} –	2.21E^{-1} –	3.95E^{-1}
MW10	4.54E^{-1} =	2.28E^{-1} –	4.37E^{-1} –	2.77E^{-1} –	2.41E^{-1} –	3.33E^{-1} –	4.54E^{-1}
MW11	4.48E^{-1} =	4.46E^{-1} –	4.45E^{-1} –	4.47E^{-1} –	4.48E^{-1} +	4.31E^{-1} –	4.47E^{-1}
MW12	6.05E^{-1} +	3.20E^{-1} –	6.01E^{-1} –	6.04E^{-1} =	6.03E^{-1} –	5.77E^{-1} –	6.05E^{-1}
MW13	4.77E^{-1} =	2.52E^{-1} –	4.64E^{-1} –	3.39E^{-1} –	3.13E^{-1} –	6.72E^{-2}-	4.76E^{-1}
MW14	4.68E^{-1} –	4.33E^{-1} –	4.67E^{-1} –	4.70E^{-1} –	4.56E^{-1} –	4.49E^{-1} –	4.70E^{-1}
+/–/=	2/1/11	0/14/0	1/11/2	5/8/1	3/11/0	0/14/0	

The symbols "–", "=", and "+" denote that the proposed algorithm is better than, similar to and inferior to the compared algorithm under the Wilcoxons rank sum test with a 0.05 significance level. The best result in each row is highlighted.

proposed algorithm explores more promising regions. Therefore, more high-quality solutions are generated, which brings better results. The performance of the proposed algorithm is similar to the MOEA/D-DAE. Both of these two algorithms can well solve the CF test suite. In terms of the DOC test suite, the proposed algorithm performs better. Compared with ToP on the DOC test suite, the proposed algorithm achieves comparable results. Since the ToP was suggested to solve the DOC test suite, this shows the practicability of the proposed algorithm in dealing with CMOPs with constraints in both the decision and objective space.

To show the effectiveness of the proposed algorithm in dealing with CMOPs with constraints in the objective space, this paper also compares it with comparison algorithms on the MW test suite. The statistical results of the average IGD and HV on MW1-14 obtained by seven CMOEAs over 20 independent runs are listed in Table 4 and Table 5, respectively. In Table 4 and Table 5, the symbol "NaN" means the corresponding algorithm fails to find any feasible solution in all 20 times runs. As shown in the two tables, the performance on the MW test suite of the proposed algorithm resembles ICMA. Because the proposed algorithm uses the same selection machine during the evolutionary process, it can also effectively find the PF. That is, it can maintain the advantages of ICMA. Therefore, the proposed algorithm performs better than other comparison algorithms on MW test suites.

4.3. Practical examples

The proposed CMOEA can be used in many practical works, such as power electronics [63], mechanical design [64], and power system optimization [65], etc. This paper considers a synchronous optimal pulse-width modulation of 3-level inverters [63], a two bar truss design [64], and an optimal droop setting [65]. The synchronous optimal pulse-width modulation of 3-level inverters problem aims to minimize the harmonic distortion and modulation index deviation under the constraints of continuity switching angles within the modulation index range associated with a given pulse number. The two bar truss design problem aims to minimize two the weight of the truss and the displacement of the joint under the constraints of the stresses induced in the members. The optimal droop setting problem aims to minimize the voltage deviation, active and reactive power loss considering the node equations for each type of bus. The detail formula of the problems are given in the appendix.

This paper runs independently 20 times of each algorithm on the practical examples. Figs. 4–5 show the performance of three best algorithms, *i.e.*, the proposed algorithm, MOEA/D-DAE and ICMA, on the examples according to the median HV value over 20 times. As shown in Fig. 4, the proposed algorithm perform better than the comparison algorithm MOEA/D-DAE on the synchronous optimal pulse-width modulation of 3-level inverters task in diversity and convergence of the obtained solutions. In addition, ICMA fails to get the feasible solutions in this task. For the two bar truss design problem, as shown in Fig. 5, three algorithms all get a great performance. As to the optimal droop setting task, Fig. 6 shows that the proposed algorithm and ICMA get the feasible solutions while MOEA/D-DAE fails to find a feasible solution. Compared with ICMA, the proposed algorithm obtain the feasible solutions with better convergence.

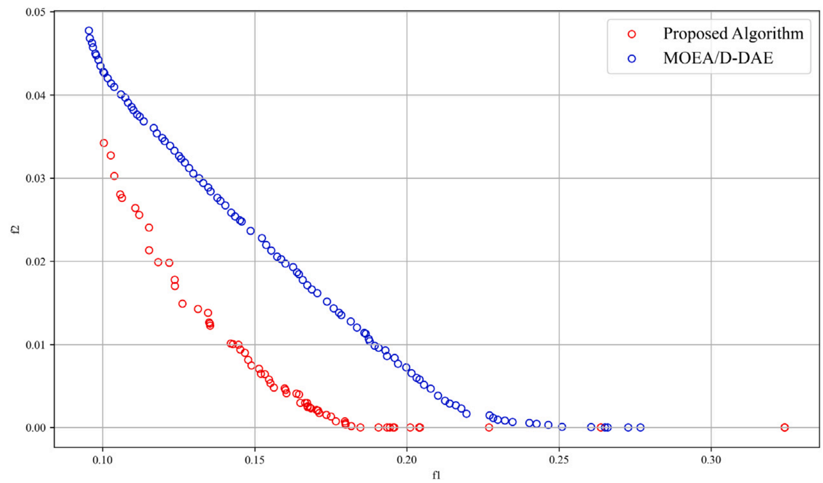

Fig. 4. The performance of the comparison algorithms on the synchronous optimal pulse-width modulation of 3-level inverters problem.

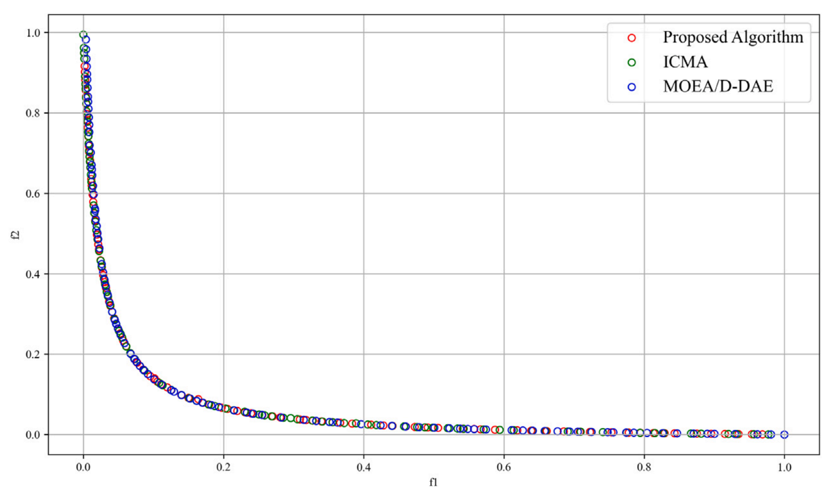

Fig. 5. The performance of the comparison algorithms on the two bar truss design problem.

5. Conclusion

This paper has proposed a new indicator to select different differential evolution operators. According to the fitness landscape and evolutionary generation, the indicator's value is determined to adapt to different CMOPs. This paper has embedded the indicator into a CMOEA and proposed a new CMOEA. The new CMOEA can balance the exploration and exploitation for solving different CMOPs with constraints in both the decision and objective space. Experiments have demonstrated the efficiency of the proposed algorithm in comparison with the state-of-the-art counterparts in dealing with CMOPs with constraints in both the decision and objective space. In addition, the proposed algorithm can perform well on CMOPs with constraints in the objective space.

CRediT authorship contribution statement

Jingjing Fang: Methodology. **Hai-Lin Liu:** Methodology, Funding acquisition. **Fangqing Gu:** Software.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Hai-Lin Liu reports financial support was provided by National Natural Science Foundation of China. Hai-Lin Liu reports financial support was provided by Programme of Science and Technology of Guangdong Province. Hai-Lin Liu reports was provided by Natural Science Foundation of Guangdong Province.

Data availability

No data was used for the research described in the article.

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Appendix A. Detail formula of the artificial example

The detail formula of each artificial example is shown in Table A.6.

Appendix B. Detail formula of the practice example.

- The synchronous optimal pulse-width modulation of 3-level inverters problem:

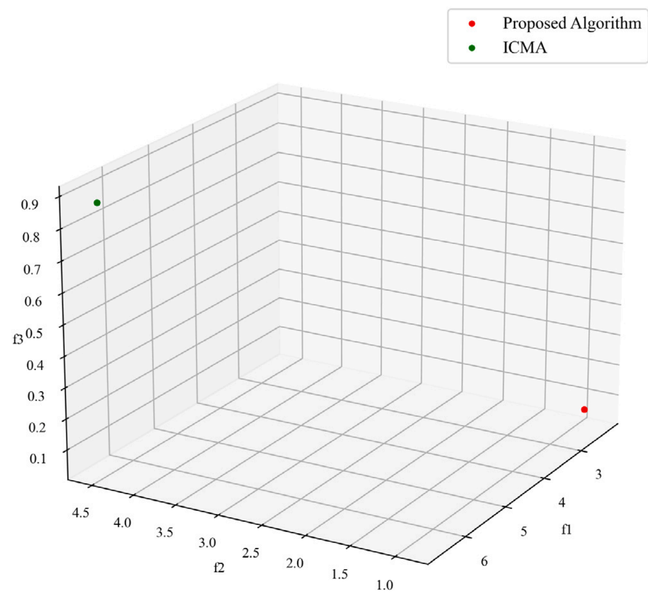

Fig. 6. The performance of the comparison algorithms on the optimal droop setting problem.

Table A.6
Detailed implementation of the proposed constraint testing problems F1-F3.

Problem	Definition
F1	$\min\begin{cases} f_1(X)=x_1\left(1+d\left(X_D\right)\right)\\ f_2(X)=\left(1-x_1\right)\left(1+d\left(X_D\right)\right)\end{cases}$ s.t. $\begin{cases} g_1(X)=x_2^2+x_3^2\le 1\\ g_2(X)\left(f_1+f_2\right)^2\le 2\end{cases}$ $d\left(X_D\right)=2\sum_{i\in D}x_i^2$, D=[2,3], $0\le x_i\le 1(i=1,2,3)$.
F2	$\min\begin{cases} f_1(X)=x_1\left(1+d\left(X_D\right)\right)\\ f_2(X)=\left(1-x_1\right)\left(1+d\left(X_D\right)\right)\end{cases}$ s.t. $\begin{cases} g_1(X)=-\sum_{i\in\{2,3\}}x_i^2+\sum_{i\in\{2,3\}}x_i\le 0.2\\ g_2(X)-\sum_{i\in\{1,2\}}f_i^2+\sum_{i\in\{1,2\}}f_i\le 0\end{cases}$ $d(X_D)=2\sum_{i\in D}x_i^2$, D=[2,3], $0\le x_i\le 1(i=1,2,3)$.
F3	$\min\begin{cases} f_1(X)=x_1\left(1+d\left(X_D\right)\right)\\ f_2(X)=(1-x_1)\left(1+d\left(X_D\right)\right)\end{cases}$ s.t. $\begin{cases} g_1(X)=-x_2^3-x_3^2\le -1 \text{ if } x_2^3+x_3^2\le 0.1\\ g_1(X)=0 \text{ otherwise}\\ g_2(X)=\left(f_1+f_2\right)^2\le 3\end{cases}$ $d(X_D)=2\sum_{i\in D}x_i^2$, D=[2,3], $0\le x_i\le 1(i=1,2,3)$.

Minimize:

$$f_1=\frac{\sqrt{\sum_k(k^{-4})(\sum_{i=1}^N s(i)cos(kx_i))^2}}{\sum_k(k^{-4})} \tag{B.1}$$

$$f_2=(m-\sum_{i=1}^N s(i)cos(x_i))^2 \tag{B.2}$$

subject to:

$$x_{i+1}-x_i-10^{-5}>0, i=1,2,\dots N-1 \tag{B.3}$$

$$0<x_i<\frac{\pi}{2}, i=1,2,\dots N \tag{B.4}$$

where x_i is the ith switching angle, $N=\lfloor\frac{Fr_{max}}{Fr_1}\rfloor$, Fr_{max} and Fr_1 are the maximum switching frequency and the fundamental frequency of the voltage waveform, respectively, odd-order harmonics $k=5,7,11,13,\dots,97$, m is the modulation index, and $s(i)=(-1)^{i+1}$.

- The two bar truss design problem:

Minimize:

$$f_1=x_1\sqrt{16+x_3^2}+x_2\sqrt{1+x_3^2} \tag{B.5}$$

$$f_2=\frac{20\sqrt{16+x_3^2}}{x_3x_1} \tag{B.6}$$

subject to:

$$f_1-0.1\le 0 \tag{B.7}$$

$$f_2-10^5\le 0 \tag{B.8}$$

$$\frac{80\sqrt{1+x_3^2}}{x_2x_3}-10^5\le 0 \tag{B.9}$$

$$10^{-5}\le x_i\le 100, i=1,2 \tag{B.10}$$

$$1\le x_1\le 3 \tag{B.11}$$

- The optimal droop setting problem:

Minimize:

$$f_1=\sum_{i=1}^N P_i \tag{B.12}$$

$$f_2=\sum_{i=1}^N Q_i \tag{B.13}$$

$$f_3=\sum_{i=1}^N(1-|V_i|^2) \tag{B.14}$$

subject to:

$$\sum_{i=1}^N(G_{ki}V_{r,i}-B_{ki}V_{m,i})-\frac{V_{r,k}P_k+V_{m,k}Q_k}{V_{r,k}^2+V_{m,k}^2}=0, k=1,\dots,N \tag{B.15}$$

$$\sum_{i=1}^N(B_{ki}V_{r,i}-G_{ki}V_{m,i})-\frac{V_{m,k}P_k-V_{r,k}Q_k}{V_{r,k}^2+V_{m,k}^2}=0, k=1,\dots,N \tag{B.16}$$

$$P_k-Cp_k(w_k^*-w)+P_{l,k}=0, k=1,\dots,N \tag{B.17}$$

$$Q_k-Cq_k(w_k^*-\sqrt{V_{r,k}^2+V_{m,k}^2})+Q_{l,k}=0, k=1,\dots,N \tag{B.18}$$

$$V_{min}\le V_{r,k},V_{m,k}\le V_{max}, k=1,\dots,N \tag{B.19}$$

$$P_{min}\le P_k\le P_{max}, k=1,\dots,N \tag{B.20}$$

$$Q_{min}\le Q_k\le Q_{max}, k=1,\dots,N \tag{B.21}$$

$$Cp_{min,k}\le Cp_k\le Cp_{max,k}, k=1,\dots,N \tag{B.22}$$

$$Cq_{min,k}\le Cq_k\le Cq_{max,k}, k=1,\dots,N \tag{B.23}$$

$$w_{min}\le w\le w_{max}, k=1,\dots,N \tag{B.24}$$

where P_i and Q_i are the active and reactive injected power at ith bus, respectively; V_i is the bus voltage at ith bus; P_k/Q_k are the specific real/reactive power injected at the kth bus, respectively; G_{ij}/B_{ij} are the real/imaginary part of i,j-th element of bus-admittance matrix, respectively; $V_{r,j}/V_{m,j}$ are the real/imaginary part of voltage of j bus, respectively; Cp_k/Cq_k are the active and reactive power droop parameters of controllers, respectively; w is operating frequency and N is the number of buses in system.

Appendix C. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.asoc.2024.112128.

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